

field by theoretically demonstrating that the electronic properties of the diode and transistor can be observed in specifically tailored optical lattices.

Researchers believe that it is possible to emulate the behaviour of a semiconductor diode in these atomic systems. For example, simulations show that this augmented optical lattice will allow atoms to flow across it from left to right, but forbids the atoms to traverse the lattice going the other way.

Ultra-cold atoms have interesting properties that conventional materials lack—superfluidity, superconductivity and coherence, to name just three. Being cold, they are also well-behaved enough to be manipulated by lasers. When several are held in a line or an array, these can link in a way that is governed by the laws of quantum mechanics and then the fun really starts.

These can be used to measure time on unimaginably short time-scales, can carry out simple calculations and may even form the basis of future quantum computers. Almost all of the atomtronics pioneers hope that for certain applications atoms will prove to be more interesting than electrons.

The motivation to construct and study atomtronic analogues of electronic systems comes from several directions:

1. The experimental atomtronic realisations promise to be extremely clean. Imperfections such as lattice defects or phonons can be completely eliminated. This allows one to study an idealised system from which all inessential complications have been stripped.

Consequently, one may obtain an improved understanding of the essential requirements that make certain electronic devices work. It is possible that a deeper understanding may provide feedback to the design of conventional electronic systems leading to future improvements. This lies parallel to the recent interest in single-electron transistors in mesoscopic systems and molecules, where many themes common with atomtronics emerge.

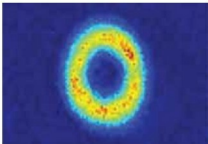


Fig. 1: Atoms spun by laser beams

A consequence of the near-ideal experimental conditions for optical lattice systems is that theoretical descriptions for atomtronic systems can be developed from first principles. This allows theorists to develop detailed models that can reliably predict the properties of devices.

2. Atomtronic systems are richer than their electronic counterparts because atoms possess more internal degrees of freedom than electrons. Atoms can be either bosons or fermions, and the interactions between these can be widely varied from short to long range and from strong to weak. This can lead to behavior that is qualitatively different to that of electronics.

Consequently, one can study repulsive, attractive or even non-interacting atoms in the same experimental set-up. Additionally, current experimental techniques allow the detection of atoms with fast, state-resolved and near-unit quantum efficiency. Thus it is possible, in principle, to follow the dynamics of an atomtronic system in real time.

3. Neutral atoms in optical lattices can be well isolated from the environment, reducing de-coherence. These combine a powerful means of state readout and preparation, with methods for entangling atoms. Such systems have all the necessary ingredients to be the building blocks of quantum signal processors. The close analogies with electronic devices can serve as a guide in the search for new quantum information architectures, including novel types of quantum logic gates that are closely tied with the conventional architecture in electronic computers.

4. Recent experiments studying

transport properties of ultra-cold atoms in optical lattices can be discussed in the context of the atomtronics framework. In particular, one can model the short-time transport properties of an optical lattice with the open quantum system formalism discussed here.

Latest developments

The atoms placed in an optical lattice, when super-cooled to form Bose-Einstein condensates, may form states analogous to electrons in solid-state crystalline media such as semiconductors. Impurity doping allows the creation of n- and p-type semiconductor analogue states, and an atomtronic battery can be created by maintaining two contacts at different chemical potentials. Analogues to diodes and transistors have also been theoretically demonstrated.

Although atomtronic devices have yet to be realised experimentally, the properties of condensed atoms offer a wide range of possible applications. The use of ultra-cold atoms allows for circuit elements, which further allow for the coherent flow of information and may be useful in connecting classical electronic devices and quantum computers.

The use of atomtronics may allow for quantum computers that work on macroscopic scales and do not require the technological precision of laser-controlled few-ion computing methods. Since the atoms are Bose condensed, they have the property of superfluidity and, therefore, have resistance-less current in which no energy is lost or heat is dissipated, similar to superconducting electronic devices. The vast knowledge of electronics may be leveraged to easily adapt to ultra-cold atomic atomtronic circuits.

Physicists have developed a new type of circuit that is little more than a puff of gas dancing in laser beams. By choreographing the atoms of the ultra-cold gas to flow as a current that can be controlled and switched on and off, the scientists have taken a step toward building the world's first 'atomtronic' device.